MAKERERE UNIVERSITY

COLLEGE OF COMPUTING & INFORMATION SCIENCES

SCHOOL OF COMPUTING & INFORMATICS TECHNOLOGY

CSC2107 DASABASE MANAGEMENT SYSTEMS
TEST 1 (Computer Science Year-2 A) 2022/2023

TIME ALLOWED: 1 Hour

Instructions

1. Attempt all questions
2. Every question should be attempted starting on a fresh page of the answer booklet.

Question One

a) With examples explain the following terms as applies to database applications development
   i. Database *(2 Marks)*
   ii. Data independence *(2 Marks)*
   iii. Data Administrator *(2 Marks)*

b) With reference to the Three-Level ANSI-SPARC Architecture, explain the difference between the external and internal levels of abstraction. *(4 Marks)*

Question Two

a) Using relevant examples, explain what you understand by the term "Data Model" as applied to databases. *(4 Marks)*

b) Briefly explain two reasons that may lead to the failure of database application development projects. *(4 Marks)*

c) Discuss what a user view represents in the context of a database system? *(2 Marks)*

Question Three

a) Briefly explain the main activities and the major output from each of the following stages of the Database Systems Development Lifecycle *(2 Marks @)*
   i. Database Planning
   ii. Systems Definition
   iii. Requirements Collection

b) State one advantage and one disadvantage of any two fact-finding techniques used during the database system development lifecycle. *(4 Marks)*

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